

JAVA EXPERT



(6 Months)

INTRODUCTION TO JAVA

- Why Java was Developed
- Application Areas of Java
- History of Java
- Platform Independency in Java
- USP of Java: Java Features
- Sun-Oracle Deal
- Different Java Platforms
- Difference between JDK, JRE, JVM
- Java Versions
- JVM Architecture
- Installing Java on Windows
- Understanding Path Variable: Why Set Path

CREATING FIRST JAVA PROGRAM

- Understanding Text Editors to Write Programs
- How to compile java file
- Byte Code and class file
- How to run class file

JAVA LANGUAGE FUNDAMENTALS

- Identifiers
- Keywords
- Variables
- Literals
- Data Types
- Operators
- Comments
- Looping Statements
- Condition Statements
- Type Casting

OOP IMPLEMENTATION (PIE)

- Why OOP
- OOP Concepts with Real life examples
- Class & its Syntax
- Object & its Syntax
- Reference Variable
- Constructors
- Instance(Non-Static) & Static Variables
- Instance(Non-Static) & Static Methods
- this Keyword and its usages
- Object & Static Initializers(Anonymous Blocks)
- Understanding '+' Operator

- Inheritance & its Syntax
- Types of Inheritance
- Object Class as Root of Java Class Hierarchy
- Variable Hiding
- Method Hiding
- Method Overriding
- Method Overloading
- Super keyword and its usages
- Final keyword and its usages
- Constructor Chaining
- Upcasting and Downcasting
- Static & Dynamic Binding
- Run Time Polymorphism
- Abstract Keyword (Abstract classes and methods)
- Understanding Interfaces
- Implementation of Encapsulation
- Association with Implementation

PACKAGES

- Understanding Packages
- Setting Class path
- Reading Input from Keyboard
- Access Modifiers
- With in Package & Outside Package Implements

NESTED TYPES

- Static Nested Class
- Non-static Nested Class
- Local Class
- Anonymous Class
- Nested Interface

ARRAYS

- General Definition of Array
- Advantages from Array
- Arrays in Java
- 1-d Arrays
- 2-d Arrays
- Jagged Arrays
- Array of reference type
- Operations on Arrays
- User Define Array & Object Type

COMMAND LINE ARGUMENTS AND WRAPPER CLASSES

- How to read command line arguments
- Wrapper Classes
- Parsing of Numeric Strings
- String representation of Primitives

EXCEPTION HANDLING

- Types of Runtime Errors
- Understanding Exceptions
- Exception Class Hierarchy
- Try & Catch Blocks
- Patterns of Catch Block
- Nested Try statements

- Throw, throws and finally
- Creating Custom Exceptions
- Checked & Unchecked Exceptions
- Assertion

WORKING WITH STRINGS

- What is String
- String Class
- Creating String Object
- Operations on String
- String Buffer Class and it's Methods
- Difference between String and StringBuffer class
- String Builder Class and it's Methods
- Difference between StringBuffer and StringBuilder

SWING

- Introduction to AWT
- Introduction to Swing Components
- Look And Feel of Swing Components
- MVC Architecture of Swing Components
- Working with Image
- Advance Swing Components
- JOptionPane,JTree,JTable,JTabbedPane
- JFileChooser,JcolorChooser
- Menu Components
- JMenu
- JMenuItem
- JMenubar

MULTITHREADED PROGRAMMING

- Multitasking: Why Concurrent Execution
- Multiprocessing v/s Multithreading
- Main Thread (Default Java Thread)
- Creating Child Threads and understanding context switching
- Thread States
- Thread Group
- Thread Synchronization: Methods and Blocks
- Inter-Thread communication
- Daemon Threads
- Deadlock

I/O STREAMS

- What is I/O
- Why Need Streams
- Byte Streams and Character Streams
- Read/Write operations with file
- Scanner Class
- Object Serialization& Deserialization
- Transient keyword
- File Class and it's Methods

SOCKET PROGRAMMING

- Understanding Fundamentals of a Network
- Socket and ServerSocket Classes
- InetAddress Class
- DatagramSocket and DatagramPacket Classes
- URL,URLConnection,HttpURLConnection Classes

REFLECTION

- Understanding the Need Of Reflection
- Getting information about class's modifiers, fields, methods, constructors and super classes
- Finding out constant and method declaration belong to an interface
- Creating an instance of the class whose name is not known until runtime
- Getting and setting values of an object's field if field name is unknown until runtime
- Invoking a method on an object if the method is unknown until runtime
- Invoking Private Methods

EXTENDED & UTILITY CONCEPTS

- Generics
- Lambda Expression
- Annotations
- Object Cloning
- Vargs
- Static-import
- Enum
- Static, Default and Private Methods of Interface
- Var Type
- Java Modules
- Stream API

COLLECTIONS FRAMEWORK

- What is Collection?
- What is Framework?
- Collections Framework
- Core Interfaces
- Collection, List, Queue, Deque
- Set, NavigableSet, SortedSet
- Map, NavigableMap, SortedMap
- Core Classes
- ArrayList, LinkedList, PriorityQueue, ArrayDeque
- HashSet, LinkedHashSet, TreeSet,
- HashMap, IdentityHashMap, WeakHashMap, LinkedHashMap, Tree Map
- Accessing a Collection via an Iterator
- Accessing List via ListIterator
- Accessing a Collection via for each loop
- Working with User Defined Objects
- The Comparator and Comparable Interfaces
- The Legacy classes and Interfaces.
- Enumeration, Vector, Stack
- Hashtable, Properties

DATE & TIME API

- java.util.Date
- java.util.Calendar
- java.sql.Date

SYSTEM PROPERTIES & INTERNATIONALIZATION (I18N)

- Understanding Locale
- Resource Bundle
- Usage of properties file
- Fetching text from Resource Bundle
- Displaying the text in HINDI
- Displaying date in Hindi

INTRODUCTION TO SQL (PROJECT BASED)

DATABASE PROGRAMMING USING JDBC

- Need Of JDBC
- JDBC Drivers
- Statement, PreparedStatement, CallableStatement
- Scrollable and Updatable ResultSet
- Batch Updates
- Transaction
- Metadata
- Connection Data Base
- Oracle
- My SQL
- Mongo DB

JAVA EE(JAVA PLATFORM ENTERPRISE EDITION)

- Understanding the Concept of Java EE : JEE Specification
- Java EE Architecture
- Single Tier
- Two Tier
- Three Tier
- N-Tier
- Java EE Components
- Web Components
- Distributed(Business) Components
- Java EE Containers& Servers
- Web Container& Web Server(Apache Tomcat)
- EJB Container& Application Server(Weblogic,Glassfish,Websphere)
- Java EE Services
- JNDI Service
- Java Transaction Service
- JAAS
- JMS

JAVA SERVLET

- Introduction to web programming
- Role of Servlet in web programming
- Servlet Lifecycle
- Servlet with Annotations
- @WebServlet
- @WebInitParam
- @WebListener
- @WebFilter
- @MultipartConfig
- Request Dispatching
- Parameters & Attributes and their differences
- ServletConfig and ServletContext
- File Uploading and Downloading
- Session Tracking&State Management
- Cookie
- Url Rewriting
- Hidden Form Field
- Session Object
- Events & Listeners

- Dependency Injection
- Refreshing Servlet
- Filters

JAVA SERVER PAGES (JSP) & JSTL

- JSP Architecture
- JSP Elements
- JSP Objects
- Understanding JavaBeans
- Custom Tags
- Using tags of JSTL
- Expression Language

PROJECT CLASSES

- Front End Coding
- FORM DESIGNING
- HTML
- CSS
- JAVA SCRIPT
- BOOTSTRAP
- Back End Coding
- DATABASE DESIGNING
- Connecting forms to database
- Writing Business Logic
- Project Hosting

DESIGN PATTERN

- Why Design Patterns...?
- Front Controller
- Composite View
- Factory Pattern
- Singleton Pattern
- DAO Pattern

JAVA MAIL API

- Email System and Protocols
- Sending & Receiving Mails
- Handling Attachments

INTRODUCTION TO DISTRIBUTED PROGRAMMING

- RMI
- Web Services

INTRODUCTION TO RESTFULL SERVICES

- @PathParam
- @Path
- @FormParam
- @QueryParam
- @DefaultValue

OVERVIEW OF JPA FRAMEWORK

SPRING

- What is Spring?
- Spring modules
- Understanding dependency Injection
- Applying aspect-oriented programming

MAVEN DEPLOYMENT

- Maven Configuration
- Converting Maven to Eclipse
- Various Maven Command

BASIC BEAN WIRING

- Containing your Bean
- Creating bean
- Injecting into bean properties
- Auto wiring
- Controlling bean creation
- Aspect Oriented Programming

HITTING THE DATABASE

- Learning spring's data Access Philosophy
- Configuring a data source
- Using JDBC with Spring
- Working with JDBC Templates
- Using Spring's DAO Support Classes for JDBC
- Integrating Hibernate with Spring
- Caching

HIBERNATE

INTRODUCTION TO ORM

- Need of ORM
- Problems using JDBC Directly
- ORM Implementation

INTRODUCTION TO HIBERNATE

- Hibernate Architecture
- Hibernate configuration
- Hibernate's Support for Other Technologies
- Installing Hibernate
- A “Hello world” stand alone application

CREATING PERSISTING CLASSES

- Mapping a basic Java Class
- Mapping a Class with Binary Data
- Mapping a Serializable Class
- Mapping a class with Data/ calendar attributes
- Mapping a Read-only class
- Mapping a class using Versioning /Timestamps

MAPPING INHERITENCE WITH JAVA CLASSES

- Table-Per –class Hierarchy Mapping
- Table-Per –subclass Hierarchy Mapping
- Table-Per –concrete-subclass Hierarchy Mapping
- Persistence interfaces

WORKING WITH COLLECTIONS

- Associations
- Lazy initialization
- Mapping Maps/Sorted Maps
- Mapping Sets/Sorted Sets
- Mapping lists
- Mapping Arrays
- Mapping a Bidirectional Association

HIBERNATE CACHING

- How caching improves performance
- First level lache
- Second level cache

SPRING BOOT

- Introduction To Spring Boot
- Spring Boot & JDBCTemplate
- Spring Boot Annotation
- Spring Boot & JPA Hibernate
- Spring Boot Rest API
- Spring Boot MVC
- Spring Boot Security
- Introduction to Micro Services



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